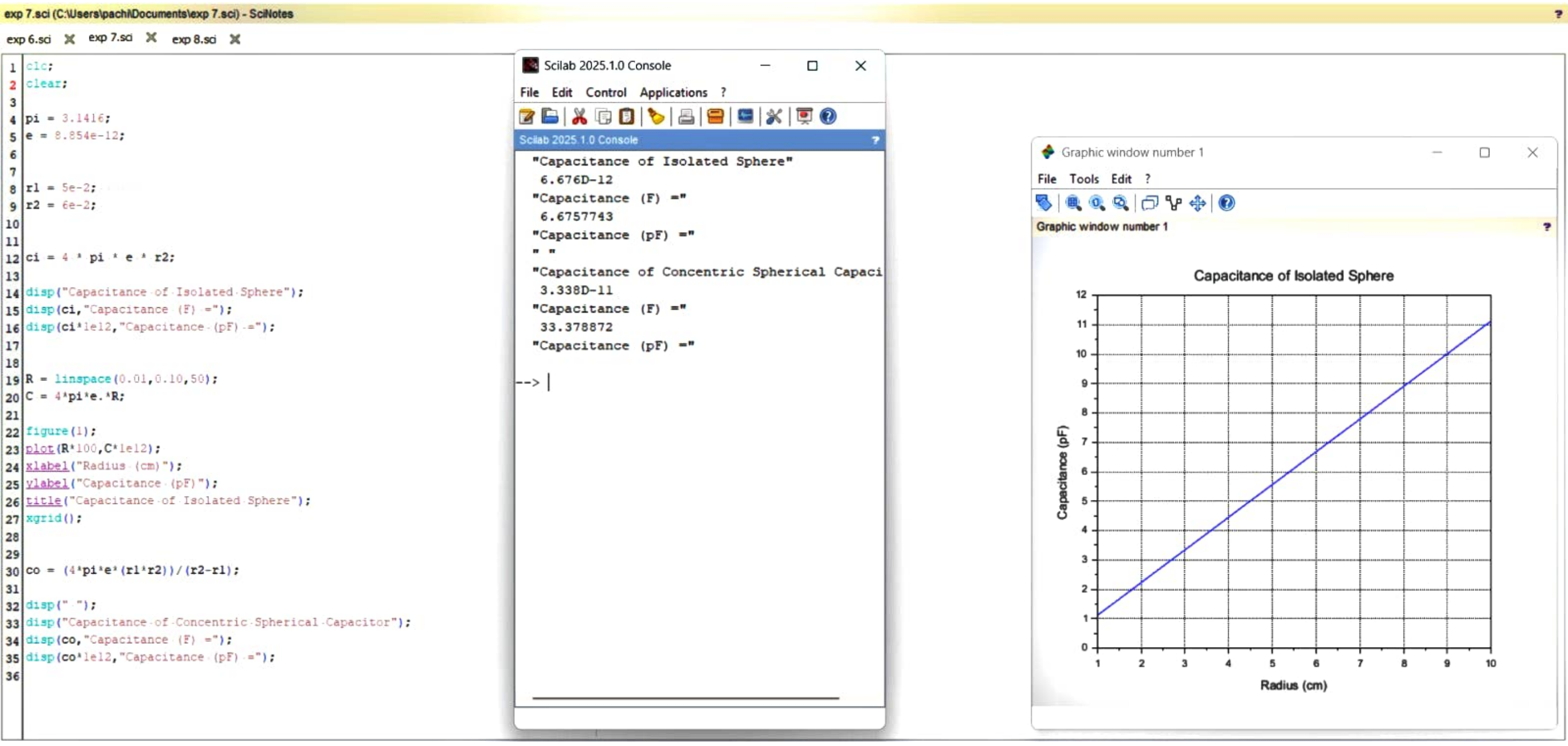

exp 7.sci (C:\Users\pachi\Documents\exp 7.sci) - SciNotes

exp 6.sci | exp 7.sci | exp 8.sci

```
clc;
clear;

pi = 3.1416;
e = 8.854e-12;


r1 = 5e-2;
r2 = 6e-2;


ci = 4 * pi * e * r2;

disp("Capacitance of Isolated Sphere");
disp(ci,"Capacitance (F) =");
disp(ci*1e12,"Capacitance (pF) =");


R = linspace(0.01,0.10,50);
C = 4*pi*e.*R;

figure(1);
plot(R*100,C*1e12);
xlabel("Radius (cm)");
ylabel("Capacitance (pF)");
title("Capacitance of Isolated Sphere");
xgrid();


co = (4*pi*e*(r1*r2))/(r2-r1);

disp(" ");
disp("Capacitance of Concentric Spherical Capacitor");
disp(co,"Capacitance (F) =");
disp(co*1e12,"Capacitance (pF) =");
```

Scilab 2025.1.0 Console

```
 "Capacitance of Isolated Sphere"
  6.676D-12
 "Capacitance (F) ="
  6.6757743
 "Capacitance (pF) ="
 " "
 "Capacitance of Concentric Spherical Capaci
  3.338D-11
 "Capacitance (F) ="
  33.378872
 "Capacitance (pF) ="

-->
```

exp-7

$r_1 = 5\ cm$

$r_2 = 6\ cm$

Capacitance of an Isolated Sphere

$$C_i = 4\pi \varepsilon_0 r_2$$

$$C_i = 4 \times 3.1416 \times 8.854 \times 10^{-12} \times 0.06$$

$$C_i = 6.677 \times 10^{-12} F$$

$$\boxed{C_i = 6.677\ pF}$$

Capacitance concentric

$$C_0 = \frac{4\pi \varepsilon_0 r_1 r_2}{r_2 - r_1}$$

O/P ✓

$$= \frac{4 \times 3.14 \times 8.854 \times 10^{-12} \times (0.05)(0.06)}{0.06 - 0.05}$$

$$= \frac{3.337 \times 10^{-13}}{0.01}$$

$$C_0 = 3.339 \times 10^{-11} F$$

$$\boxed{C_0 = 33.39\ pF}$$